

Slide 1 – Title Slide

Speaker Notes:

- Good afternoon, everyone. I'm Long Yin Chu, and this presentation covers my dissertation work on "Development and Evaluation of a Low-Cost, EMG-Controlled Prosthetic Arm Using 3D Printing Technologies." Dr. James May supervised this project.
 - Over the next 15 minutes, I will walk you through:
 1. Why this problem matters (motivation & context)
 2. Why we chose EMG sensing combined with 3D printing
 3. Our aims & objectives
 4. How the EMG collects signals
 5. Firmware design for real-time control
 6. Mechanical design highlights and assembly
 7. Testing results, discussion, and limitations
 8. Conclusions and future work
 - By the end, you will see how we built a functional prosthetic arm for under £100, what performance metrics it meets, and why a low-cost, open-source approach can make prosthetics accessible in low-resource settings.
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Slide 2 – Motivation & Context

(Original "Problem Overview" / "Motivation & Context")

Speaker Notes:

1. **Global Need + Cost Barrier**
 - Over **30 million people** worldwide require a prosthetic or orthotic device, yet **75 percent** in developing regions lack access. The main bottleneck is cost (tens of thousands of pounds) and the need for specialized fabrication (e.g., machining centers, injection molding).

- In high-income countries, top-tier myoelectric hands (e.g., Ottobock Michelangelo, i-Limb Quantum) cost **£60 000–£100 000**. Even “low-cost” mid-tier arms like Open Bionics’ Hero Arm (£7 600) or Otto Bock’s Zeus Arm (£6 000) remain beyond reach for most users in lower-income settings.

2. Abandonment & Quality-of-Life Impact

- Because of these prohibitive costs, nearly **half of upper-limb amputees** abandon their prostheses within one year—either due to unaffordable maintenance or inadequate functionality for daily tasks.
- Without even a basic functional hand, amputees face severe limitations in independence, employment opportunities, social participation, and overall mental well-being.

3. Project Goal

- We aim to **bridge the cost-performance gap**. Our objective: build a basic yet functional, EMG-controlled prosthetic arm for **under £100** (parts cost) by leveraging low-cost electronics and 3D printing.
- The end goal is a **fully open-source**, easily reproducible design that clinics, NGOs, and community makerspaces can replicate and adapt in low-resource environments.

Slide 3 – Why EMG + 3D Printing?

Speaker Notes:

1. EMG Sensing: Low-Cost, Non-Invasive Control

- **Surface EMG (Electromyography)** measures the small voltage changes (50 μ V–a few mV) generated by muscle fibers during contraction. We place two Ag/AgCl electrodes over the biceps ($\approx$ 2 cm apart) and a reference electrode on a nearby bone (e.g., lateral epicondyle) to reject common-mode noise.
- Because those raw signals sit at tens to hundreds of microvolts, we must amplify, rectify, and filter them to create a reliable “muscle-intensity” envelope the Arduino can read. Once processed, an Arduino Nano can sample and classify “open” vs. “close” commands in near real time.
- **Key advantages:**
 - Non-invasive (no surgical implants)
 - Very low hardware cost (INA333 or AD8226 + TL084 op-amp chain instead

of a \$200-\$500 commercial EMG module)

- Rapid response (<300 ms end-to-end) yields acceptable real-time control for simple grasps

2. 3D Printing (FDM with PLA): Democratized Fabrication

- **PLA filament** costs ~£20/kg; each arm uses 200 g (£4). No CNC or injection molds are required.
- We designed all finger segments, palm, wrist module, and forearm housing in CAD (SolidWorks & Fusion 360) for **tool-free, snap-fit** assembly and minimal supports during printing.
- **Benefits of 3D printing:**
 - Local, decentralized manufacturing: any FDM printer (costing <£300) can replicate the parts.
 - Rapid iteration: a design change can be printed and tested overnight.
 - Modular & maintainable: uniform finger segments, a removable palm lid, and “pop-off” forearm panels let users repair or replace components in under 15 minutes.
- **Cost comparison:**
 - University-printed PLA parts: near £0 (in-house) vs. outsourced quoting ~£253.
 - Combined with cheap electronics (<£100 total), we drastically undercut even the “mid-tier” 3D-printed arms on the market, which often still rely on fiberglass sockets or water-jet-cut brackets.

3. Synergy: EMG + 3D Print

- Merging **low-cost EMG control** with **3D printed mechanics** produces a prosthesis that retains core functionality (finger flex, wrist rotation) at a fraction of commercial pricing.
- By open-sourcing CAD files, schematics, and firmware, we empower communities to **adapt** geometry (e.g., resize for a child’s limb) or swap materials (e.g., PETG, TPU) as needed.

Slide 4 – Aims & Objectives

Speaker Notes:

1. Primary Aim

- Develop a **modular**, EMG-controlled prosthetic arm (hand + wrist) using Arduino-compatible electronics and desktop FDM 3D printing, at a **parts cost under £100**.

2. Functional Objectives

- **Wrist Articulation:** Yaw ($\pm 12.5^\circ$), pitch ($\pm 35^\circ$), and **full 360° roll** to let the user orient tools or cups.
- **Finger Flexion/Extension:** One DOF per digit: thumb + four fingers on a shared tendon-pulley system.
- **Real-Time Control:** End-to-end latency < 300 ms so users perceive the hand as “responsive.”
- **Grip Force:** Target ~ 6 N combined (sufficient to lift a 500 mL water bottle or small tool).

3. Engineering Objectives

- **Electronics:** Use an Arduino Nano and a simple analog EMG front-end (AD8226 $\rightarrow$ TL084 rectifier & filter) for minimal cost and easy maintenance.
- **CAD & Mechanical:** Design all parts for tool-free, “snap-fit” assembly—focus on reinforcing joints to minimize tendon slack and finger misalignment.
- **Power Architecture:** Isolate EMG rails (± 9 V batteries) from servo rails (6 V Li-Po) to prevent analog noise and servo brownouts.

4. Constraints

- **Budget:** Strictly “all electronics + mechanical materials” $\leq$ £100 (excluding in-house 3D print labor).
- **Simplicity:** Non-expert end users in low-resource settings must be able to assemble, calibrate, and maintain it.
- **Scope:** Exclude an elbow joint in this version (drives up cost & complexity). Focus on **below-elbow** prosthesis (wrist + hand).

Slide 5 – How the EMG Signals Work

Speaker Notes:

1. Underlying Physiology (Action Potentials & Summation)

- When you voluntarily flex your biceps, **motor neurons** trigger a chain of action potentials in muscle fibers. Those thousands of fibers fire nearly simultaneously, and their small voltage changes ($\approx 50 \mu\text{V}$ –few mV) **sum together** at the skin surface.
- **Surface EMG electrodes** “listen” to that combined signal, but it arrives as a **biphasic** waveform (positive + negative swings around 0 V) with frequencies up to several hundred hertz. The raw amplitude is too small to feed directly into a microcontroller.

2. Electrode Configuration & Placement

- Two **Ag/AgCl** desktop electrodes are spaced ~ 2 cm apart over the biceps belly (midpoint between tendon insertion and muscle belly). A third electrode (reference) sits on a bony landmark (e.g., lateral epicondyle) to reject noise.
- This **differential arrangement** ensures we see only the true muscle voltage, while canceling common-mode interference (e.g., 50 Hz mains hum, cross-talk from nearby muscles).

3. Signal Detection

- Because the raw signal is on the order of **tens to hundreds of microvolts**, we need an **instrumentation amplifier** to boost it into the millivolt range without introducing noise. Our final schematic uses an **AD8226** (an INA333 alternative) with a gain resistor $R_G = 240 \Omega$, giving about **206 $\times$ gain**. That transforms a $50 \mu\text{V}$ spike to ≈ 10 mV.

4. Key Takeaway

- At this stage, we capture the muscle’s biphasic EMG spikes (± 10 mV after amplification), but they are still too “jittery” to directly command a hand. We must convert them into a smooth unipolar “envelope” that corresponds to overall contraction intensity.

Slide 6 – EMG Front-End: Amplification $\mapsto$ Rectification $\mapsto$ Filtering $\mapsto$ DC Offset

Speaker Notes:

1. **Amplification (AD8226, $\sim 206\times$ Gain)**

- The **AD8226** instrumentation amplifier uses a single $R_G = 240\Omega$. $G = 1 + (49.4k\Omega / 240\Omega) \approx 206 \times$. $G \approx 1 + (49.4k\Omega / 240\Omega) \approx 206 \times$.
- This high gain ensures that even a small $50\mu V$ action potential becomes ≈ 10 mV. The AD8226's high CMRR (> 90 dB) rejects mains interference and cable noise.
- **Result:** A clean millivolt-level bipolar waveform centered around 0 V.

2. Full-Wave Rectification (TL084 Precision)

- A simple diode bridge would drop ~ 0.7 V and destroy small peaks. Instead, we use a **TL084 op-amp "super-diode"** configuration: two TL084 stages invert any negative half-cycle, then sum them to produce only positive voltages.
- Output after rectification: a unipolar signal (0–20 mV) that preserves the amplitude of every muscle spike.

3. Low-Pass Filtering (TL084, 80.6 k Ω - 1 μF , $f_c \approx 2$ Hz)

- Even the rectified waveform is full of high-frequency ripples from individual muscle fibers (~ 50 – 500 Hz). We want just the **envelope** (0–2 Hz) that correlates with contraction strength.
- We employ a **first-order active filter**: $R_{13} = 80.6k\Omega$, $C_1 = 1\mu F$. $f_c = 1 / (2\pi \cdot 80.6k\Omega \cdot 1\mu F) \approx 1.97$ Hz.
- That yields $f_c = 1 / (2\pi \cdot 80.6k\Omega \cdot 1\mu F) \approx 1.97$ Hz and a time constant $\tau \approx 0.08$ s.
- **Effect:** Rapid spikes above 2 Hz are attenuated; the output is a smooth DC-like voltage that rises as the user tenses and falls as they relax. There is ~ 180 ms of latency in the envelope itself, but this remains within our < 300 ms latency budget when combined with the Arduino's processing.

4. DC Offset (2.5 V Bias for Arduino Compatibility)

- The Arduino Nano's ADC can only measure **0–5 V**. After filtering, the envelope sits at 0–20 mV around 0 V, which we cannot feed directly. So we add a **2.5 V DC bias** using a resistor divider and summing amplifier stage.
- **Outcome:**
 - At rest (no contraction), the filtered output sits at ≈ 2.5 V at the Arduino pin.

- At full contraction, the envelope might rise +100 mV, so the Arduino sees $\approx 2.6 \text{ V}$.
- In firmware, we subtract 2.5 V to recover only the muscle intensity portion ($\pm 0.1 \text{ V}$), which maps to a handful of ADC counts—enough to compute a reliable RMS threshold.

Slide 7 – Arduino RMS Processing & Control Algorithm

Speaker Notes:

1. Sampling & Buffering

- The Arduino Nano's **analogRead(A0)** samples the DC-shifted EMG envelope at **10 Hz** (every 100 ms). Each raw reading (0–1023 counts for 0–5 V) is stored in a **circular buffer** of length $N=50$ samples.

2. RMS Computation

- At each loop iteration:

$$\text{RMS} = \sqrt{\frac{1}{N} \sum_{i=1}^N (V_i - 2.5)^2}$$
 where $V_i = (\text{analogCount}_i / 1023) \times 5 \text{ V}$

$$\text{RMS} = \sqrt{\frac{1}{N} \sum_{i=1}^N ((\text{analogCount}_i / 1023) \times 5 - 2.5)^2}$$
- Effectively, we compute a **running-window RMS** over the last 5 seconds of data (50 samples at 10 Hz).

3. Dynamic Baseline & Hysteresis Thresholding

- We maintain a second buffer of the last **30 RMS values** (~3 s of data). The average of those 30 values is the user's **baseline** RMS (representing “resting tone + noise”).
- We define two thresholds:
 - **“Tense”** → when $\text{currentRMS} > 1.10 \times \text{baseline}$
 - **“Relax”** → when $\text{currentRMS} < 0.98 \times \text{baseline}$

- The **2 percent gap** between “tense” and “relax” prevents rapid toggling (chattering) if the RMS hovers near threshold.

4. State Machine & Servo Commands

- Two states: `STATE_OPEN` (hand open) and `STATE_CLOSED` (hand closed). Initially, the system starts in `STATE_OPEN`.
- If `state == STATE_OPEN AND currentRMS > 1.10 × baseline`, switch to `STATE_CLOSED` → `servoAll.write(0°)` (close hand).
- If `state == STATE_CLOSED AND currentRMS < 0.98 × baseline`, switch to `STATE_OPEN` → `servoAll.write(180°)` (open hand).
- After issuing a servo command, we **ignore** further RMS changes for ~250 ms (blocking code) to give the servos time to move and avoid command overlap.

5. Performance Metrics from Trials

- **End-to-End Latency:** We timestamp the moment RMS crosses the “tense” threshold and timestamp when `servo.write(0°)` is called. Over five cycles, we observed **274 ms, 279 ms, 288 ms, 268 ms, 264 ms** → **avg = 275 ms** ($\sigma \approx 8.8$ ms).
- **Classification Accuracy:** In a short 5-cycle “open→close” test, we recorded 3 True Positives (correct closes), 1 False Negative (missed tension), and 1 False Positive (spurious closing). That yields **80 percent accuracy**.
- **Key Insight:** A simple RMS + dynamic threshold algorithm provides **80 – 90 percent** reliability at minimal cost, without machine learning.

Slide 8 – Hardware & Assembly – Servo Pulley System

Speaker Notes:

1. Servo Selection

- We use three **Tower Pro MG995** hobby servos ($\approx$ £15 each). Each is rated at ~10 kg·cm torque at 6 V and draws 250 mA – 2.5 A (stall).
- **Mapping of Servos to Digits:**

- **Servo 1:** Drives **thumb + index** (via a small PLA “Y-splitter” on the pulley horn).
- **Servo 2:** Drives **middle + ring** (another Y-splitter).
- **Servo 3:** Drives **pinky** + an extra tendon that wraps around the wrist assembly to control **360° roll**.

2. Tendon-Driven Pulley Design

- We use **high-tensile nylon fishing line** (0.5 mm diameter) as tendon. Each finger’s tendon runs from the fingertip, through the palm’s internal channel, over the ball joint in the wrist, then straight to the corresponding servo horn in the forearm.
- **Flexion/Extension:** Clockwise horn rotation pulls the tendon for flexion. Counter-clockwise rotation releases it, allowing the finger’s passive elastic rebound to extend back to open.
- **Friction & Slack Losses:**
 - **PLA ↔ nylon contact:** ~25 percent of tension is lost to friction along each 50 mm segment.
 - **Shared ring + pinky routing:** Adds ~15 percent extra slack, reducing force for those digits.
- We incorporate **tuning bars** under the palm lid so that users can adjust tendon tension in the field if slack builds up.

3. Theoretical Finger Force Calculation

- Each MG995 servo: stall torque ~0.98 N·m at 6 V; horn radius ≈ 10 mm → tendon tension ≈ 98 N.
- For a fully extended finger, distance from MCP joint to fingertip ≈ 65 mm, and tendon anchor sits ~4.5 mm from MCP.

$$F_{\text{fingertip}} = \tau_{\text{servo}} \times \frac{r_{\text{horn}}}{d_{\text{anchor}}} = 0.98 \text{ N}\cdot\text{m} \times \frac{0.01 \text{ m}}{0.065 \text{ m}} \approx 1.51 \text{ N}.$$

$$F_{\text{fingertip}} = d_{\text{anchor}} \times \frac{\tau_{\text{servo}}}{r_{\text{horn}}} = 0.065 \text{ m} \times \frac{0.98 \text{ N}\cdot\text{m}}{0.01 \text{ m}} \approx 1.51 \text{ N}.$$
- When the finger is bent (~35 mm anchor to fingertip),

$$F_{\text{fingertip}} = 0.98 \text{ N}\cdot\text{m} \times \frac{0.01 \text{ m}}{0.035 \text{ m}} \approx 2.80 \text{ N}.$$

$$F_{\text{fingertip}} = 0.035 \text{ m} \times \frac{0.98 \text{ N}\cdot\text{m}}{0.01 \text{ m}} \approx 2.80 \text{ N}.$$
- **Ideal combined force (excluding thumb):** ~1.5 N + 2.8 N for each of four fingers ≈ 17 N total. In reality, friction & slack reduce usable force to ~1.0 –

1.5 N per digit (≈ 6 N combined), which matches our kitchen-scale measurements.

4. Finger Actuation Speed

- The MG995 moves **60° in ~0.16 s**. For $\sim 170^\circ$ of flexion (≈ 3 cm tendon travel), that equates to **~0.46 s** per full open→close cycle. This is slightly slower than commercial devices but acceptable for basic grasp tasks.

5. Takeaway

- Although we lose ~ 75 percent of theoretical force to friction/slack, we still achieve ~ 6 N combined grip—sufficient to lift light objects (cups, phones).
- The modular pulley design allows easy maintenance: if a tendon breaks, a user can re-thread it through the palm lid in under 5 minutes without disassembling the entire wrist or forearm.

Slide 9 – System Diagram & Power Architecture

Speaker Notes:

1. System Block Diagram (show figure on slide)

- **EMG Electrodes** → differential leads → **AD8226 Amp** → **TL084 Rectifier** → **TL084 LPF + DC Bias** → **Arduino A0**
- **Arduino Nano** (RMS + threshold logic) → **PWM pins** → **MG995 Servos** → **Tendon Driven Hand & Wrist**

2. Power Supplies

- **EMG Front-End:** Two 9 V batteries in series to provide ± 9 V rails for the AD8226 and TL084. This separation keeps analog rails quiet by isolating them from the noisy servo currents.
- **Arduino Nano:** Powered from a regulated **5 V USB** supply (a small 5 V buck module steps down the 7.4 V Li-Po). This ensures the microcontroller never takes noise from the servo bus.
- **Servos ($\times 3$ MG995):** Powered by a **7.4 V Li-Po (2000 mAh)** through a 6 V BEC (battery-elimination circuit). Each servo draws ~ 0.25 A idle, ~ 0.5 A under moderate load, and ≤ 1.2 A stall. Combined, moderate use is ~ 1.5 A–2 A,

giving ~1 – 1.5 h run time.

3. Grounding & Isolation

- All grounds (EMG pack, Arduino ground, servo ground) converge at a **single star point** inside the forearm housing. This prevents ground loops, which could otherwise inject noise into the analog signal.
- We route the EMG wires twisted together for the first 50 mm to reduce EMI pickup. Power lines (servo + battery) are physically separated from EMG analog traces by at least 5 mm.

4. Runtime Estimates

- **Servos:** At average 1.5 A draw (moderate grasp cycles), a 2000 mAh pack yields ~1.3 h. In practice, idle times between grasps extend that to ~1.8 h.
- **EMG Module:** Two 9 V alkalines last ~2 h of continuous amplifier operation before sagging below ± 8 V (which starts to degrade common-mode range). Rechargeable 9 V Li-Ion would be lighter and more reliable long-term.

5. Key Takeaway

- By keeping analog and digital power rails separate, we ensure a stable EMG front-end. If we had tied everything to the same battery, servo spikes would garble the EMG signal and make thresholding unreliable.

Slide 10 – Mechanical Design – Finger & Palm

Speaker Notes:

1. Finger Design Iterations

- **Iteration 1 (Baseline):**
 - Three identical PLA segments per finger (proximal, middle, distal) printed flat. We cut a 1.2 mm channel through each for the fishing-line tendon, plus a V-groove to allow flexion.
 - **Issue:** Under tension, finger segments misaligned by 2–3 mm because the V-grooves lacked lateral constraints, leading to inconsistent grip.

- **Iteration 2 (Reinforced Joints):**
 - Inserted an **M1.5 steel pin** through each hinge eyelet so rotation occurs around the pin instead of friction between PLA. This prevented mid-segment skew.
 - Added **snap-fit fingertip caps** to seal the tendon channel and keep the line centered.
 - Standardized segment geometry across all five fingers (so one sliced STL file prints all 15 segments in one batch).
- **Outcome:**
 - Joint stiffness improved by ~30 percent, yielding consistent finger alignment under load.
 - Tendon stayed centered, reducing slack by ~2 mm.

2. Palm Design Iterations

- **Iteration 1 (Simple Cavity):**
 - Anatomically curved palm with internal cavities for tendon routing, but walls only 2 mm thick, which flexed and caused tendon rubbing. Tendon exit holes (1.2 mm) were too tight, creating spikes in friction.
- **Iteration 2 (Reinforced Lid + Channels):**
 - Increased wall thickness to 3 mm, added localized gussets around tendon exits for mechanical support.
 - Introduced a **removable palm lid** (secured by two M2 screws) that exposes all tendon anchor points—so users can fine-tune tension without fully disassembling the hand.
 - Reoriented the thumb anchor to a natural 45° opposition angle.
- **Outcome:**
 - Ease of assembly/maintenance: field adjustment of tendon tension in under 10 minutes.
 - Robustness: thicker walls eliminated unwanted flex under a 70 N tendon pull.

3. Key Takeaways (Finger + Palm):

- **Tendon Slack Is Critical:** Even a single millimeter of misalignment drastically reduces grip force. Reinforcing joints and sealing channels fixed that.
 - **Modularity & Field Repair:** Standardized finger segments mean if any segment cracks, you print one new piece and pop it in—no need to reprint the entire hand.
 - **Print Orientation:** We print finger segments lying flat to avoid warping. If printed vertically, each layer's cross-section shifts, leading to “fat” hinges that don't fit.
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Slide 11 – Mechanical Design – Wrist & Forearm Housing

Speaker Notes:

1. Wrist Module Iterations

- **Iteration 1 (InMoov-Style Gears):**
 - Tried an open-source design with interlocking bevel and spur gears to control yaw, pitch, and roll. That required about 12 printed gear pieces, precise tolerancing (± 0.05 mm), and ~4 hours of combined print time. In practice, small print errors caused jamming and backlash.
- **Iteration 2 (Ball Joint + Tendon Anchors):**
 - Switched to a **20 mm PLA sphere** in a **22 mm socket**. This ball joint gives friction coupling in all orientations.
 - **Roll (360°):** A single tendon anchored to the bottom of the sphere runs to servo 3. Rotating the sphere around its axis gives full 360° rotation.
 - **Yaw ($\pm 12.5^\circ$) & Pitch ($\pm 35^\circ$):** Two small tendon anchors on the “equator” of the sphere: pulling on the “north” anchor pitches up; pulling on the “east” anchor yaws right.
 - Added an “A-frame” limit stop on the socket to prevent over-rotation ($> \pm 45^\circ$ in any direction), protecting tendons from overextension.
- **Outcome:**

- Simplified assembly: only a single printed sphere + socket (no gears).
- Print-time reduced from ~4 hours to ~1.5 hours (socket + sphere printed together).
- Tolerant of ± 0.2 mm print errors.

2. Forearm Housing

- **Iteration 1 (Segmented Cuff):**
 - Initially tried a segmented design where the electronics and servos slide into a stack of rings (like InMoov). In practice, clips were fragile and wiring got pinched between segments.
- **Iteration 2 (Cylindrical Shell + Removable Panels):**
 - Designed a single hollow cylinder with three **removable rectangular panels** (secured by M2 thumb screws).
 - **Internal Mounts:**
 - **Arduino Nano + Breadboard:** Snap-fit bosses align with nylon standoffs—no wobble.
 - **Servo Brackets:** Three angled servo mounts at the bottom align horns with tendon anchor posts.
 - **Wiring Channels:** Integrated ribs keep loose wires away from spinning horns. A small clip holds the EMG ± 9 V battery pack.
 - If a servo fails, the user simply removes one panel and swaps it out—no need to disassemble the entire wrist or hand.
- **Outcome:**
 - Field-serviceable: repair or swap a servo in under 5 minutes.
 - Single-print parts minimize assembly complexity—no separate inserts or fragile clips.

3. Design Philosophy

- **Minimal Supports:** Every part is oriented so that its largest flat face sits on the print bed—no supports except a 1 mm brim for the sphere-in-socket.
- **Tolerance Compensation:** All mating surfaces have a 0.3 mm clearance (e.g., sphere vs. socket, servo horns vs. tendon posts) to account for PLA shrinkage.

Slide 12 – User Experience, Accessibility & Sustainability

Speaker Notes:

1. Ease of Assembly & Use

- **Step-by-Step Build Guide:**

- Print all PLA parts (fingers, palm, wrist, forearm panels) on any standard FDM printer (PLA, 0.2 mm layer, 20 percent infill).
- Assemble finger segments with M1.5 pins, thread fishing-line tendon, snap on fingertip caps.
- Mount three MG995 servos in forearm housing, wire to Arduino Nano and separate 6 V battery.
- Build EMG front-end on a small perf-board (AD8226 + TL084), place electrodes on biceps (clean skin with alcohol), adjust offset.
- Upload Arduino firmware, hold muscle relaxed for 5 s to calibrate dynamic baseline.
- Strap on the arm (Velcro socket or generic bracket), attach palm, route tendons, and insert palm lid.
- Test open/close; if slack arises, tighten tendon via tuning bars under the palm lid.

- **User Learning Curve:**

- The only tricky part is **correct electrode placement** (≈ 2 cm below deltoid insertion). With annotated photos in the guide, a novice can learn this in < 10 minutes.
- Once calibrated, it is **plug-and-play**: flex biceps to close, relax to open—no ML training required.

2. Accessibility & Local Manufacturing

- **3D Printer Availability:**

- FDM machines (costing < £300) are now common in universities, NGOs, and makerspaces. PLA filament (< £20/kg) is widely available.
- Any community workshop can print 10-20 arms per month with 2–3 printers.
- **Open-Source DIY Kit Potential:**
 - By publishing all CAD files (STL/STEP), firmware, circuit schematics, and a detailed **Bill of Materials (BOM)** on GitHub, local clinics or tech schools worldwide can produce, customize, and iterate.
 - Adjust finger lengths, switch to PETG or TPU for durability, or even embed a simple sensor in the palm for haptic feedback.

3. Sustainability

- **PLA's Pros & Cons:**
 - PLA is biodegradable under industrial conditions but does not readily break down in a home compost. It's ideal for low-temperature, low-warp printing.
 - We recommend collecting failed parts for local recycling if facilities exist.
- **Repairability:**
 - Individual finger segment cracks can be fixed by printing and replacing a single segment (10 minutes print time).
 - Tendon breaks require < 5 minutes to re-thread.
 - Servo replacement is a **three-screw** job via the forearm's removable panels.

4. Societal Impact & Ethical Considerations

- **Societal Impact:**
 - A functional prosthetic for < **£100** (versus £6 000–£100 000) drastically improves quality of life for amputees in low-resource areas: restoring basic grasp, building confidence, enabling employment.
 - By empowering local technicians to maintain and adapt the design, we foster **ownership** and **dignity**.

- **Ethical & Safety Considerations:**
 - Entirely **non-invasive** EMG (no personal data stored). For any clinical rollout, we must pursue **FDA/CE** approval, conduct rigorous long-term safety, comfort, and durability testing (e.g., repeated stress cycles, user comfort surveys).
 - We must also ensure **universal design**: adjustable sockets (e.g., thermoplastic liners) to accommodate different residual limb shapes, including pediatric or geriatric populations.
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Slide 13 – Testing & Evaluation Metrics

Speaker Notes:

1. Quantitative Metrics

- **Latency (ms):** Time from EMG threshold crossing → first observable finger motion. Target < 300 ms.
- **Range of Motion (ROM):**
 - **Fingers:** Measured with a plastic goniometer:
 - Index: 61.2°
 - Middle: 59.8°
 - Ring: 48.5°
 - Pinky: 44.3°
 - Thumb: 46.5°
 - **Wrist:**
 - Yaw: ± 12.5°
 - Pitch: ± 35°
 - Roll: 360° continuous
- **Grip Strength (N):** Individual fingertip forces (measured on a kitchen scale):
 - Index: 1.42 N
 - Middle: 1.50 N
 - Ring: 0.98 N
 - Pinky: 0.97 N
 - Thumb: 1.32 N
 - **Combined:** ≈ 6 N (sufficient to lift light containers).
- **Control Accuracy (%):**
 - 5-cycle “open→close” test: 3 True Positives, 1 False Negative, 1 False Positive → **80 percent**.

- **Power & Runtime:**
 - Servos draw ~1.5 A under moderate load → 2000 mAh Li-Po gives ~1–1.5 h of typical use.
 - EMG ±9 V batteries last ~2 h before sagging.
- **Cost & Build Time:**
 - Total parts cost ~£91 (rigidly under £100).
 - CAD design & mechanical iteration: ~120 hours over 8 weeks.
 - Circuit breadboarding & tuning: ~15 hours (1 week).
 - Firmware + debugging: ~8 hours.
 - Integration & testing: ~15 hours (1 week).
 - **Total labor:** ~158 hours (~4 weeks full-time equivalent).

2. Measurement Methods

- **Latency:** Two Arduino pins toggle on threshold crossing and on servo write. Logged on a two-channel oscilloscope; average ≈ 275 ms (range 264 – 288 ms).
- **ROM:** Goniometer measurements for each finger and digital inclinometer for wrist.
- **Grip Strength:** Fingertip pressed on kitchen scale; average of three trials per finger.
- **Control Accuracy:** Logged every “StateChange” event during a 5-cycle user test; classified manually.
- **Power:** Inline current sensor logged servo bus current every 5 s for 10 minutes of continuous cycles. Extrapolated capacity.
- **Cost/Time:** Detailed spreadsheet of invoices and timesheets tracked in Excel.

3. Key Takeaway

- Our prototype meets the **sub-£100** cost goal, achieves ≈ **275 ms** latency, ≈ **6 N** combined grip, and **80 percent** classification accuracy—demonstrating feasibility of a **low-cost**, functional, EMG-driven prosthesis.

Slide 14 – Discussion – Comparative Analysis

Speaker Notes:

1. Latency & Responsiveness

- **Prototype:** Average end-to-end latency ~ 275 ms (100 ms sampling + 100 ms RMS buffer + 75 ms servo movement).
- **Commercial Mid-Tier (e.g., Hero Arm, i-Limb):** Use dedicated EMG front ends + DSP, achieving ~ 100–150 ms latency.
- **Impact:** > 250 ms starts feeling “laggy” in rapid tasks. Ours is slightly above that threshold but remains acceptable for basic pick-and-place. If we increase sampling to 20 Hz and reduce RMS window to 30 samples (1.5 s window at 20 Hz), we could shave ~100 ms—at cost of more RMS jitter.

2. Range of Motion & Grip Strength

- **Prototype ROM:**
 - Fingers: 44–61° flexion vs. human max ~75–90° → ~ 65–85 percent of natural flex on index/middle. Sufficient for a power grasp on common objects.
 - Wrist: $\pm 12.5^\circ$ yaw, $\pm 35^\circ$ pitch vs. $\pm 45^\circ$ in many mid-range arms. Roll: 360° matches or exceeds some.
- **Commercial Devices:**
 - Hero Arm offers 14 unique grip patterns with finger flexion ~ 90° and grip force ~ 15–20 N.
 - i-Limb Quantum can generate ~ 30 N total grip force with 24 grip patterns.
- **Implication:**
 - We trade off **fine dexterity** (e.g., independent pinky movement, multi-digit pinch) and raw strength for cost. But for a user who has never had any functioning prosthesis, our “**good-enough**” 60° flexion and ~ 6 N grip is already life-changing.

3. Control Accuracy

- **Prototype:** 80 percent (binary tense/relax) using a single-channel RMS threshold.
- **Commercial Pattern Recognition (e.g., Coapt, MyoPro):** 8–16 channels, machine learning classifiers (LDA/SVM), > 95 percent accuracy across multiple gestures—requires 30 minutes of training per user.
- **Trade-off:**
 - Our approach: **zero training**, a single electrode → “flex biceps to close, relax to open.” Achieves ~ 80 percent out of the box.
 - Upgrading to multi-channel ML would raise cost by at least £20–£40 and require user training.

4. Cost-Performance Trade-off

- **Prototype (< £100):** Basic open/close grasp and wrist rotation; modules are field-repairable.
 - **Commercial Devices (£6 000–£100 000):** High-fidelity multi-pattern grasp, greater grip force, medical-grade materials for durability (silicone gloves, carbon fiber, etc.).
 - **Key Insight:** For a user in a low-resource setting who never had any prosthesis, our “barebone” functionality (grasp key objects, rotate a cup, open a door) is far better than no hand at all—while reducing hardware costs by > 95 percent compared to mid-tier devices.
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Slide 15 – Discussion – Innovation & Limitations

Speaker Notes:

1. Innovative Elements

- **Modular CAD-Optimized Design:** Snap-fit finger segments, removable palm lid, friction ball-joint wrist → field reparability.
- **Low-Cost EMG Front-End:** Using an **AD8226** for ~206× gain + TL084 rectifier/filter yields a stable envelope without an expensive commercial EMG board.
- **Simple Arduino RMS Algorithm:** No machine learning—just a 50-sample RMS window + dynamic baseline threshold yields ~80 percent accuracy.
- **Hybrid Tendon & Pulley:** Integrated PLA channels plus fishing-line tendon deliver decent force at minimal cost; tuning bars allow in-field slack adjustment.
- **Open-Source Philosophy:** Entire BOM, CAD, and firmware published—enables global replication and iterative improvement.

2. Limitations & Bottlenecks

- **Mechanical Slack & Friction:**
 - Shared ring + pinky routing introduced ~15 percent extra slack.
 - PLA ↔ nylon friction (~25 percent loss) plus a 0.3 mm clearance at each joint → ~75 percent total force loss from theoretical torque.
 - **Fix:** Use independent Dyneema cords with PTFE liners, or embed

low-friction bearings.

- **Servo Torque Decay:**
 - 7.4 V Li-Po under load sags to ~5.5 V, reducing MG995 torque to ~68 percent of rated (10 kg·cm → ~6.8 kg·cm).
 - **Fix:** Use a stable 6 V BEC or a higher-voltage servo + step-down regulator.
- **Limited Fine Motor Control:**
 - Only **binary** open/close (no proportional grip or multi-pattern). Adding more gestures would require extra EMG channels + firmware complexity.
- **No Elbow Joint:**
 - Without an articulated elbow, reach envelope is limited. Adding an elbow servo + potentiometer encoder would add ~£30 – £40 and ≥ 200 g weight.
- **Durability (PLA vs. TPU):**
 - PLA finger joints begin to micro-crack after ~2000 cycles under ~3 N loads.
 - **Fix:** Switch finger joints to TPU or carbon-fiber PLA for ~10× fatigue life.
- **Control Accuracy (80 percent vs. > 95 percent):**
 - Single-channel RMS is susceptible to electrode slip or weak contractions. A second EMG channel (triceps) or a lightweight ML model on a Nano 33 BLE could boost accuracy to > 95 percent.

3. User Feedback (Informal Trials)

- **Volunteer 1:** “It responds well, but when I try to gently grasp a paper cup, it sometimes closes too quickly.” → Suggest adding a small “beep” so the user hears once the threshold is crossed.
- **Volunteer 2:** “My skin gets sweaty, so dry electrodes slip.” → Medical-grade gel electrodes or MyoWare sensors would help maintain stable contact.
- **Volunteer 3:** “I worry about long-term durability of PLA joints.” → Confirmed by our fatigue tests; TPU substitution is next.

4. Takeaway

- This prototype is a **proof-of-concept**: it demonstrates that a functional, EMG-controlled arm can be built for < £100. It is not a final commercial product, but it lays the groundwork for **iterative improvement** by the community (e.g., e-NABLE style) to refine materials, add proportional control, or embed an elbow joint.
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Slide 16 – Conclusions & Future Work

Speaker Notes:

1. Key Achievements

- Delivered a **modular**, EMG-driven prosthetic arm (hand + wrist) for **under £100** in parts.
- Achieved:
 - **Latency:** ~275 ms
 - **Grip Force:** ~6 N combined
 - **Classification Accuracy:** ~80 percent (five-cycle test)
 - **ROM:** Index 61.2°, Middle 59.8°, Ring 48.5°, Pinky 44.3°, Thumb 46.5°; Wrist yaw $\pm 12.5^\circ$, pitch $\pm 35^\circ$, roll 360° .
- Produced a fully **open-source** repository with CAD files, firmware, circuit schematics, BOM, and assembly guide.

2. Near-Term Upgrades (Low-Cost, High-Impact)

- **Improve Tendon System:** Switch from fishing line to **Dyneema** fiber + PTFE tubing to reduce friction by ~25–40% and lower slack.
- **Stabilize Power Supply:** Introduce a **6 V BEC** or higher-voltage servo + step-down regulator to consistently deliver the MG995's rated 10 kg·cm torque.
- **Hardware RMS Tweak:** Increase Arduino sampling to 20 Hz and reduce RMS buffer to 30 samples—shaving ~100 ms from latency (at cost of slightly more noise).
- **Material Upgrade:** Print finger joints in **TPU** or carbon-fiber PLA to increase fatigue life from ~2 weeks to ~2 months under daily use.

3. Mid-Term Enhancements (Moderate Cost)

- **Additional EMG Channel:** Add a second MyoWare channel on **triceps** (~£20). This allows co-contraction detection (pinch vs. power grasp) and could raise classification accuracy to ~90 percent.
- **Simple Gear Train:** Integrate a small 2:1 spur gear for finger tendon pulleys → reduce required servo torque, increase transmitted force from ~25 percent efficient to ~45 percent.

4. Long-Term Innovations (Higher Cost & Complexity)

- **Machine-Learning EMG Classification:**
 - Switch to an **Arduino Nano 33 BLE** (built-in AI accelerator) and collect multi-channel EMG data (biceps, forearm flexors, extensors).
 - Train a tiny CNN or LDA classifier to recognize 4–6 gestures (open, close, pinch, wrist flex, wrist extend) at > 95 percent accuracy—adds ~£15 hardware + training time.
- **Servo-Driven Elbow Joint:**
 - Add a high-torque micro-servo (e.g., MG996R + 3:1 spur gear) at the socket interface to provide ~100° of elbow flexion with a potentiometer for position feedback. Adds ~£30 – £40 in cost and ~200 g weight.
- **Clinical Trials & Extended Ergonomics:**
 - Partner with a local hospital or NGO for a **5–10 participant user study**, measuring Activities of Daily Living (ADL) performance (e.g., Box & Blocks Test, Jebsen Hand Function Test).
 - Collect long-term feedback on **socket comfort**, weight distribution, and user satisfaction to refine the design for real-world adoption.
- **Expand Open-Source Ecosystem:**
 - Host a GitHub repository with detailed **Assembly Videos**, multi-language documentation, and community forums. Encourage contributions from makers worldwide (e-NABLE style).

5. Final Thought

- By prioritizing **affordability**, **modularity**, and **open-source collaboration**, this project takes a meaningful step toward democratizing prosthetics. Even if it cannot match high-end commercial arms on dexterity or durability, it provides “**good-enough**” functionality to restore independence and dignity to amputees who otherwise would have no access at all.

Slide 17 – Questions

Speaker Notes:

- Thank you for your attention. I'd be happy to take any questions—whether on the EMG front-end design, 3D printing choices, firmware implementation, performance results, or next steps.